

Effects of using head-up display in automobile context on attention demand and driving performance

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Abstract

This study aimed to investigate the difference in driving performance between drivers' attention on the head-up display (HUD)/road under low/high road conditions via a driving simulator experiment. Experimental driving included four driving scenarios with attention-on-the-HUD followed by attention-on-the-road or vice versa under high or low driving load conditions. Each scenario took about a 30-min driving consisting of two 15-min sections for each attention location. Forty-eight participants, divided into four groups, drove one of the four scenarios once. Besides driving safely within speed limit, participants were also required to perform detection task and speed limit sign response task. Results revealed that drivers paying attention to the HUD, under both low and high driving load conditions, reacted faster to speed limit sign changes than when paying attention to the road. In addition, attention-to-the-HUD under low driving load condition caused the smallest variation in steering wheel angle and lateral acceleration. These differences can be attributed to the driver's enhanced awareness and the cognitive capture effect, and tended to diminish with increasing driving workload. Finally, attention shift of drivers and the so-called novelty effect for using new technology product were also found.

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Keywords: Attention shift; Cognitive capture; Driving simulator; Head-up display; Novelty effect

1. Introduction

In recent years, with the advances in global position system (GPS) and communication technology, automobile manufacturers in Taiwan have begun to produce vehicles that are capable of presenting information to the driver by way of head-up or head-down displays (HUD or HDD) (e.g. Nissan Toobe[®] and Ford e-Car[®]). While drivers can benefit from receiving immediate information concerning vehicles status and navigation, they have to face heavier driving load at the same time. Automobiles from the above-mentioned manufacturers are usually equipped with a 6–8 in. LCD installed near the air-conditioner or the stereo control panel, which is known as the HDD for presenting related information.

Summarizing previous relevant studies that drivers mostly use (over 90%) their vision in obtaining related driving information [1]. Therefore, while

the above-mentioned display systems can provide convenience and immediate information on the one hand, these systems tend to add extra load to the drivers' visual systems on which they depend heavily. In addition, when receiving related information provided by the HDD, drivers need to keep their eyes off the road, which would seriously influence driving safety [2]. Distraction from the road condition is one of the main causes of traffic dangers [3,4].

On the other hand, HUD can reduce the frequency and duration of the driver's eyes-off-the-road by projecting the required information in front of the driver. This enables the driver to steer easily [5,6] and to respond quickly to information provided by the road conditions and communication systems [7,8]. Given the advantages that it can help reduce the driver's need to shift attention from the road in front to the related visual display interface and the time needed for re-accommodation [8–11], HUD seems to be a feasible and superior alternative or auxiliary visual display interface.

The first operational HUD was flown in the Hawker-Siddeley Buccaneer in 1960. Having been used in aviation

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context for over 40 years, HUD has been employed in many studies on aviation safety. Although some of these results can be taken as reference for this research, due consideration should be given to the differences between the much more complicated road traffic conditions and the wide open sky [5,12].

Most research on HUD used as in-vehicle information display interface focused on how drivers abide by the speed limit restriction and how they react to the sudden changes in road conditions. Compared with traditional HDD users, drivers using HUD can better detect the abrupt changes on the road and maintain more stable driving speed (5 mile above or below the speed limit). In addition, drivers on average spend less time looking at the HUD than checking the HDD [13–15]. Details of related studies on specification design of HUD for automobile information display, such as size, font type, location and viewing angle can be found in some technical reports [16,17].

Given the complicated road traffic conditions, car drivers are usually viewing a rapidly changing background. Hence, objects in the environment may exert some accommodative pull on users of automotive HUD who need to focus simultaneously objects at different depths in the visual field. Study showed that when drivers attend to the information on the HUD, the objects close to it will have a so-called ‘shrink’ effect, i.e. compared with the information displayed on the HUD, the size of automobiles or pedestrians in a very close distance will become smaller [18]. In this way, drivers will overestimate the distance between them. In a study using a driving simulator, he found that while attending to the information on the HUD, drivers tend to steer into a congested traffic environment unnoticeably [19].

Besides the potential negative influences mentioned above, there are also doubts on whether the HUD can really serve its function in reducing the time needed to shift attention from the road when attending to information displayed in front, which in turn, enables the driver to make swifter and more appropriate response. In addition, there is the so-called cognitive capture effect [20], which denotes the inefficient attention switching from HUD to primary tasks. Such inefficiency will result in missing external targets or delayed responses. Given that previous research has obtained positive evaluation on traditional HDD in terms of accuracy in receiving displayed information [21] and faster reaction time [22,23], whether the HUD is superior to the traditional HDD in enhancing driving safety remains to be explored.

To assess the feasibility of HUD as an alternative in-vehicle visual display, this study aims to examine two fundamental questions. First, are there any differences in drivers’ performance between their attending to the HUD and focusing on the road? Second, are there any differences in performance due to the drivers’ attention switch from the HUD to the road and vice versa?

2. Methods

2.1. Participants

Forty-eight students from the National Yunlin University of Science and Technology participated in the study. They were divided into four groups with equal number of males and females in each group. All subjects were licensed drivers with either normal or corrected-to-normal vision. They all passed the Ishihara Test for Color Blindness and had normal hearing ability, meaning they can carry on a conversation easily with the experimenter while the simulator engine is running. In addition, they should at least drive twice a week and have had no experience with the driving simulator and the HUD. Each participant was given a cash reward of US \$10 upon completion of the test.

2.2. Apparatus and displays

2.2.1. Driving simulator

This study used the interactive STI low cost, fixed-base driving simulator developed by Systems Technology, Inc., Hawthorne, CA, USA. The STI driving simulator has been used for evaluating human driving performance and investigating medical impairment under various conditions. In the present study, the simulated vehicle cab, a VOLVO DL340, included all the normal automotive displays and control found in an automatic vehicle.

2.2.2. Head-up display

Fig. 1 shows a layout sketch of the simulator, screen and the HUD. With reference to the HUD used in previous studies [16,24–26], a HUD with the same functions and that can communicate with the STI simulator used in this research was built using the same image projection principles. In addition, according to the recommendations [17], the HUD should be located between 2.5 and 4 m and positioned from 4 to 12° below the drivers’ horizontal viewing line. In this way, the image of the HUD in this research was projected 3.1 m in front of the driver, the vertical projecting angle is between 6 and 12° below the driver’s horizontal visual line, and the HUD area is about 32(w) × 22(h) cm² (~15 in.). Other related descriptions are: display resolution 800 × 600 dpi, presentation font (icon) size 10 × 10 cm² (~1.8°).

2.3. Driving loads

Driving load should have a significant influence on the driver’s reactions when using HUD [27,28]. Therefore, the driving conditions in this research were divided into two levels, low and high loads [29]. The factors to be considered are shown in Table 1.

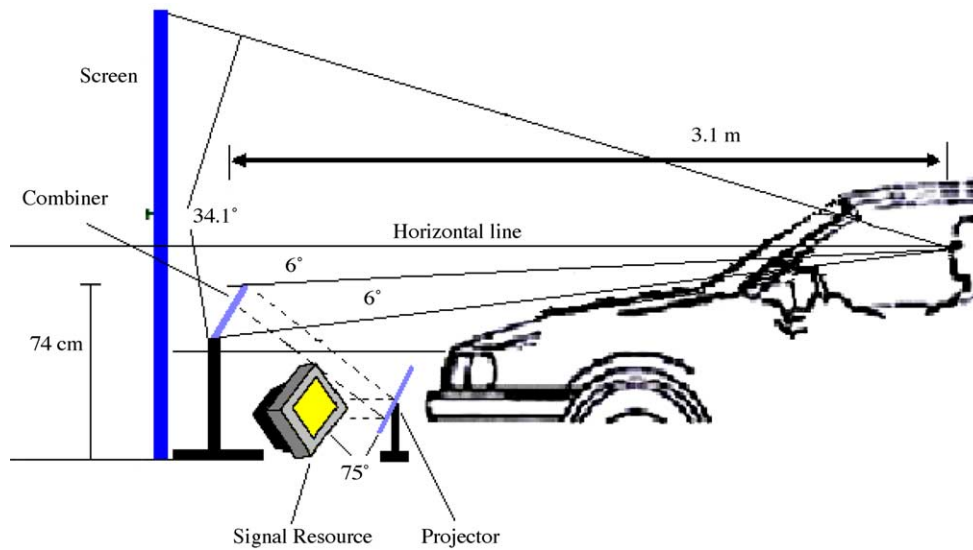


Fig. 1. A layout sketch of the simulator, screen, and the head-up display.

2.4. Driver's attention locations and their sequence

To examine whether there are any differences in drivers' performance between their attending to the HUD and focusing on the road, the driver's attention locations were manipulated as follows. First, a series of numbers (0–9) appeared at random within an interval of about 2 s, and a prompt letter (A) appears every 20 s or so. When the prompt letter vanishes from sight, a detection task follows, which is to be performed by the participant within 2 s (Section 2.5.2). If the above-mentioned numbers and the prompt letter appear at the right side of the road, the attention of the participant, theoretically, should focus on the road and herein refers to 'attention on the road' (Fig. 2(a)). On the other hand, when they are displayed on the HUD, the participants will focus their attention on the HUD and herein referred to as 'attention on the HUD' (Fig. 2(b)).

In addition, for a new technology product, i.e., the HUD, users might show a so-called 'novelty effect' [5]. Novelty effect may cause first-time users to perform poorer

than those with prior experience. Therefore, the drivers' attention locations were arranged as attention-on-the-HUD followed by attention-on-the-road or vice versa.

2.5. Tasks

While following all traffic rules and complying with safety precautions, participants were required to perform the following two tasks as fast as possible and making as few mistakes as possible.

2.5.1. Speed limit sign response task

Participants were instructed to apply force onto the gas/brake pedal in response to the different speed limit signs (i.e. 55 and 65 mph) that appeared randomly and alternatively at the side of the road or on the HUD with approximately equal frequency (around every 2 min).

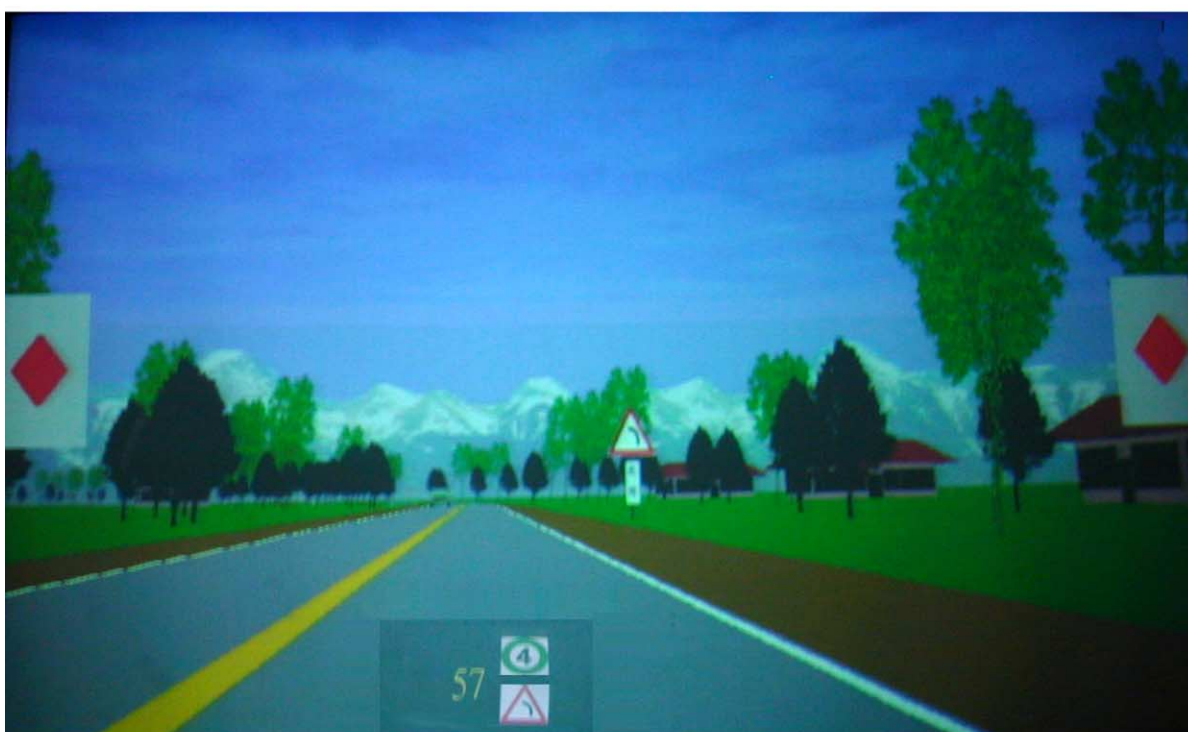
Note that there exists some discrepancy between the speed limit sign presented on the HUD and that on the roadside. That is, the sign appearing on the HUD will

Table 1
Load factors for low and high driving loads

Load factors	Driving load levels	
	Low	High
Lane width (m)	4.1	3.6
Road type	Straight two way lane	Curvy two way lane
Number of easy curves	5 (3100 m radius)	5 (3100 m radius)
Number of sharp curve	0	5 (1500 m radius)
Speed limits	88 km/h (55 mph), 104 km/h (65 mph)	88 km/h (55 mph), 104 km/h (65 mph)
Density of oncoming vehicles	Low (one vehicle per 550 m in average)	High (five vehicles per 100 m in average)
Number of intersection	28 in average	80 in average
Density of roadside building	Low (2 buildings in every 2 min driving)	High (20 buildings in every minute driving)
Location of roadside building	20 m away from the roadside	3 m away from the road side



(a)



(b)

Fig. 2. Examples of manipulating drivers' attention locations: (a) attention on the road, (b) attention on the HUD.

always look the same, while that appearing on the roadside will have its size changing from small to large. This study conducted a pilot study with subjects reading the speed limit signs appearing from far to near with the vehicle moving at

a constant speed of 55/65 mph maintained by the simulator. The results showed that the average distance for the sign appearing on the roadside to be seen clearly (error-free) was around 5 s. Coincidentally, this value approximates

the amount of time needed to make a centring corrective manoeuvre during lane keeping as found by previous study [30]. According to this, speed limit sign presented on the roadside should appear 5 s ahead of the approaching driver, and the sign appearing on the HUD should also remain on display for 5 s. However, the vehicle speedometer was always displayed on the head-up position.

There are two purposes for designing this task. One is to evaluate the differences in situational awareness of the drivers' responses in detecting those unanticipated signs appearing suddenly when their attention locations are different. The other is to understand further the differences in the drivers' attention switch requirements when their attention locations are not the same as where the sign appeared (i.e. drivers' attention was on the HUD, and the speed limit sign appeared at the roadside, or vice versa).

2.5.2. Detection task

About 2 s after the above prompt letter A disappeared, a red diamond-shaped figure appeared at the left or right side of the road randomly in a fixed position as shown in Fig. 2. Participants were required to detect this figure and quickly signal the corresponding left or right turn. The figure disappeared after 2 s. This task was designed to investigate the difference in response for different drivers' attention locations when there is a pre-alert cue for performing an oncoming event.

2.6. Scenario descriptions

The driving scenarios required for this study were prepared with STI scenario control language (SDL Version 8.1). Each participant took a 3-min practice before the actual experimental driving. Experimental driving included four driving scenarios with attention-on-the-HUD followed by attention-on-the-road or vice versa under high- or low-driving load conditions. Each scenario took about 30 min of driving within the speed limit, and consisted of two 15-min sections for each of the two attention locations. Participants were told to bring the car to a complete stop after finishing the first driving section to switch position of attention. They were allowed a short rest in between the two sections. For the speed limit response task, the speed limit information in each section will appear randomly with similar frequency at the right side of the road or on the HUD.

2.7. Experimental designs

This research was of a 2 (driving load, between-subjects) $\times$ 2 (attention location, within-subject) $\times$ 2 (sequence of driver's attention location, between-subjects) mixed factorial designs. Both objective and subjective data were collected. The driver's steering behaviours (i.e. mean speed, mean throttle/brake forces, variation in

longitudinal/lateral acceleration and steering wheel angle, and average lateral lane position and its variation), response times and error rates when performing the above two tasks (i.e. response in detecting the speed limit changes, and the red diamond-shaped figures) were collected by the simulator as objective data. The response time is defined as the time between the appearance of the speed limit sign and when force applied by the participant in response exceeded 10% of the original. If the speed was to be increased from 55 to 65 mph, the force exerted on the accelerator needed to exceed 10% of the original. In the case when the speed was to be reduced, the force applied on depressing the braking pedal should be increased by 10%. The response time of the detection task was defined as the time between the appearance of the diamond-shaped figure and when the participant indicated the corresponding turn signal. The subjective workload and preferences were collected using a Likert 5-point scale questionnaire, with 1 indicating 'very low/dislike it very much', and 5, 'very high/like it very much'. The questions probe into the subjects' perception of visual effort: amount of visual scanning required; time pressure: amount of time available for completion of the driving and the related tasks; psychological stress: feelings of frustration, confusion, confliction, and anxiety, as well as preferences during and upon completion of each scenario.

Collected data were analysed with ANOVAs via SPSS Version 10.0, significant level was specified as $\alpha < 0.01$, and post hoc analyses were carried out by Scheffé method.

3. Results

This research had collected a huge amount of data on drivers' performance. However, only topics of interests in this study and the statistically significant analysis results will be described in this section. In addition, the significant results of distinct behaviours will be interpreted by looking for supporting evidence across all the performance measures collected.

Table 2 displays the behaviours of drivers under different driving load conditions and attention locations. As can be seen, when the driving load was low, attention-on-the-HUD will enable the drivers to respond promptly to speed limit signs and perform correct changes in driving behaviours (i.e. lateral acceleration, steering wheel angle). That is to say, the response time under the attention-on-the-HUD scenario was shorter and there was less variance in movement than under the attention-on-the-road scenario. On the other hand, when the driving load was high, the response to speed limit signs when focusing on the HUD was still faster, but the changes in the vehicle's lateral acceleration were more variable than when the drivers' attention was mostly on the road.

Table 3 depicts the analysis results of driving performances vs. the shifts of drivers' attention under different

Table 2

Performance measures for attention-on-the-HUD vs. attention-on-the-road under different driving load conditions

Driving loads	Performance measures	Attention on the road	Attention on the HUD	<i>F</i> (1, 44)	<i>P</i> value
Simple	Mean RT for speed limit sign detection (s)	3.3469	1.7126	33.669	0.0001
	Variance in lateral acceleration (ft/s ²)	6.3313	3.3841	15.801	0.0001
	Variance in steering wheel (degree)	3.1665	1.5349	21.918	0.0001
Complex	Mean RT for speed limit sign detection (s)	3.2389	1.7246	45.670	0.0001
	Acceleration variance (ft/s ²)	2.7923	3.3394	7.202	0.008

driving loads when the drivers' attention was mostly on the HUD (or road) with driving-related sign information presented on the road (or HUD). The post-hoc analysis revealed consistent results regardless of the driving load and the initial attention location. In face of unanticipated speed limit signs, the driver's response time was faster when they appeared on the HUD than when they appeared by the road. Under low driving load, if the signs appeared on the HUD, whether the driver focused his/her attention on the HUD or on the road, the control of the automobile was comparatively smooth, as evidenced by small variations in steering wheel angle. However, if the driver had his/her attention on the HUD while the signs appeared by the roadside, then his/her control of steering wheel control would be at its worst.

Another topic to be investigated in this research was whether the drivers' performances will be affected by the sequence of attention location. The analysis results were given in Table 4. As can be seen, under both low and high driving loads, performance, in terms of faster response times for both test tasks and smaller variation in lateral acceleration, was better when attention was first on the road than when attention was later on the road. However, the reverse is true for the attention-on-the-HUD scenario. That is, drivers had better driving behaviours, in terms shorter response time and smaller variation in lateral acceleration and steering wheel angle, when attention was later on the HUD than when attention was first on the HUD. In addition, the drivers' mental pressure load measures as reflected by their frustration ratings also showed a similar result.

Nevertheless, when the driving load became high, difference in evaluation on mental pressure load was not significant.

Comparison between behaviours resulting from different attention location sequences reveals that under low driving load, performed measures for the vehicle control behaviours (i.e., variance in steering wheel angle and mean lateral lane position) and driver's attention demands (i.e. variance in lateral acceleration) were the best when their attention was focused later on the HUD, followed by attention-first-on-the-HUD. Conversely, attention-on-the-road, whether first or later, resulted in inferior performance.

On the other hand, under high driving load, the above-mentioned differences were reduced. Attention-later-on-the-HUD and attention-first-on-the-road resulted in similar performance in detection tasks and response time to unanticipated speed limit signs; and both were better than that under the other two scenarios, i.e. attention-first-on-the-HUD and attention-later-on-the-road.

Irrespective of the attention location sequence, response time to sudden condition (speed limit signs) was faster when the drivers focused on the HUD than on the road. Notably, under both low and high driving loads, drivers responded the slowest under the attention-later-on-the-road scenario. When driving load is low, attention-first-on-the-HUD and attention-later-on-the-road had similar psychological workload ratings. They both caused greater frustration than the other two situations, attention-first-on-the-HUD and attention-later-on-the-road.

Table 3

Performance measures for drivers' attention-on-the-HUD vs. attention-on-the-road with sign information presented on different locations and under different driving loads

Driving loads	Performance measures	Drivers' attention on the road		Drivers' attention on the HUD		<i>F</i> (3, 138)	<i>P</i> value
		Sign information presented on the road	Sign information presented on the HUD	Sign information presented on the road	Sign info. presented on the HUD		
Low	Mean RT for speed limit sign detection (s)	3.7784 B	1.3725 A	3.6029 B	1.3651 A	27.459	0.0001
	Variance in steering wheel (degree)	2.8291 BC	1.5078 A	3.1450 C	1.9210 AB	4.585	0.004
High	Mean RT for speed limit sign detection (s)	3.2822 B	1.6286 A	3.1955 B	1.8205 A	15.244	0.0001

Table 4
Performance measures for drivers' attention location sequences under different driving load conditions

Driving loads	Performance measures	Attention first on the road	Attention later on the road	Attention first on the HUD	Attention later on the HUD	<i>F</i> (3, 44)	<i>P</i> value
Low	Mean lateral lane position (ft)	6.7653 AB	6.4297 A	6.3998 A	6.9781 B	6.270	0.0001
	Mean RT for detection task (s)	0.7031 A	0.9327 B	0.8598 B	0.6713 A	10.257	0.0001
	Mean RT for speed limit sign detection (s)	2.7320 B	3.9619 C	1.3819 A	2.0432 AB	16.207	0.0001
	Variance in lateral acceleration (ft/s ²)	6.1365 B	6.5257 B	4.5969 B	2.1712 A	7.231	0.0001
	Variance in steering wheel angle (degrees)	3.1095 C	3.2235 C	2.0348 B	1.0350 A	8.797	0.0001
	Frustration ratings	1.75 A	2.9167 B	2.9167 B	2.33 AB	4.481	0.008
	Mean RT for detection task (s)	0.6935 A	1.0606 B	0.9890 B	0.6603 A	20.833	0.0001
High	Mean RT for speed limit sign detection (s)	2.6140 B	3.8638 C	1.6689 A	1.7803 A	22.054	0.0001
	Variance in lateral lane position (ft)	2.3277 A	3.3428 B	3.5297 B	2.3423 A	9.394	0.0001

4. Discussion and conclusions

To sum up briefly, use of HUD can enable drivers to respond faster to unanticipated road events (i.e. speed-limit detection and response tasks designed for this study) under both low and high driving loads. Furthermore, under low driving load, drivers have improved driving behaviours (Table 2) as evidenced by smaller variances in lateral acceleration and steering wheel angle. These valid indicators of required attention for driving [31,32] show that drivers need to pay less attention when using the HUD. As seen in Table 1, under low driving load, less attention required may reduce the overall awareness of drivers [29], thereby encouraging careless or reckless driving. If this is the case, drivers' better performance when paying attention to the HUD can mostly be attributed to their enhanced awareness. From this, it can also be inferred that the use of HUD will increase the driver's mental load to some extent. This inference can be supported by the frustration ratings shown in Table 4 (HUD vs. Road: 2.6234 vs. 2.3334 on an average). The relatively relaxed state of the driver when focusing on the road will change to that of caution when the driving load increases, therefore, under high driving load, the above-mentioned deterioration in driver's performance (i.e. variances in lateral acceleration and steering wheel angle) does not occur. However, the added load due to the use of HUD does not impose any notable negative effect on driving under such conditions. Nevertheless, these results do not agree with those obtained by previous studies [21,28]. They found that HUD when used for high workload aviation will result in longer response time.

The switch of driver's attention from HUD to road (or vice versa) has a different impact on driving behaviours. Regardless of the initial attention location (HUD/road), the performance of drivers is always the best when the attention

is shifted to the HUD. On the contrary, performance deteriorates when the attention is shifted from HUD to the road (Table 3). However, our findings are not consistent with those of McCann et al. [20]. Appearance of related information on the HUD (especially the information that requires a quick response) seems to enable the driver to have a quicker response time. The poorer performance of the driver when shifting attention from HUD to road (Table 3) may be attributed to the 'salient effect' produced by the contrast/transparency and lower visual realism of non-conformal image displayed on the HUD and of the related scenes around the road (Fig. 2), and thus, the cognitive capture effect [20,28] is also found, as evidenced by the degradation of responses to external targets due to processing of information displayed on the HUD. Designers of HUD should take the cognitive capture effect into consideration as to reduce its potential risks.

This research also reveals that the use of advanced technology products (e.g. HUD) may result in a 'novelty' effect [5]. In other words, the behaviour of first-time users may not be as good as those with prior experience (Table 4), implying that to realize the advantages of advanced technology may require previous training and adaptation. There will be no improvement in driving due to the use of HUD if the driver is not familiar with it. As seen in Table 4, performance under attention-first-on-the-HUD is worse than that under attention-first-on-the road. Deterioration in behavioural response whilst focusing on the road may, to some extent, be attributed to drivers' fatigue (Table 4).

In order to get the driver to focus on the road, numbers (0–9) and prompt character (A) are arranged at the right side of the road. To some extent, the driver's attention is thus limited to the right of the road. Though all regulatory traffic signs in Taiwan are located on the right side of the road, this cannot fully simulate real driving conditions. As a result,

there may be some errors in the evaluation. In addition, despite the significant results listed in the tables, when looking at the individual values (e.g. variance in steering wheel angle 3.2235 for attention-later-on-the-road and frustration ratings in Table 3), even the worse behavioural performances still meet the safety requirements of normal driving.

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